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Multi-member frame assembly for a track system

Abstract

A multi-member frame assembly for a track system includes a pivot pin defining a pivot axis, a leading frame member pivotally connected to the pivot pin for pivoting about the pivot axis, and a trailing frame member pivotally connected to the pivot pin for pivoting about the pivot axis, the trailing frame member pivoting independently from the leading frame member. A resilient member is pivotally connected to the leading frame member and to the trailing frame member. The resilient member extends generally vertically between the leading frame member and the trailing frame member. A track system having a multi-member frame assembly is also described.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a National Stage entry of International Application No. PCT/IB2021/050506, under 35 U.S.C. § 371, filed Jan. 22, 2021, which claims priority to U.S. Patent Application Ser. No. 62/964,959 entitled “Track System” and filed on Jan. 23, 2020, which is incorporated herein by reference in its entirety.

FIELD OF THE TECHNOLOGY

(1) This present technology relates to frame assemblies for track systems.

BACKGROUND

(2) Certain vehicles, such as, for example, agricultural vehicles (e.g., harvesters, combines, tractors, agriculture implement, etc.) and construction vehicles (e.g., bulldozers, front-end loaders, etc.), are used to perform work on ground surfaces that are soft, slippery and/or uneven (e.g., soil, mud, sand, ice, snow, etc.).

(3) Conventionally, such vehicles have had large wheels with tires to move the vehicle along the ground surface. Under some conditions, such tires may have poor traction on some ground surfaces and, as these vehicles are generally heavy, the tires may compact the ground surface in an undesirable way owing to the weight of the vehicle. As an example, when the vehicle is an agricultural vehicle, the tires may compact the soil in such a way as to undesirably inhibit the growth of crops. In order to reduce the aforementioned drawbacks, to increase traction and to distribute the weight of the vehicle over a larger area on the ground surface, track systems were developed to be used in place of at least some of the wheels and tires on the vehicles.

(4) Despite ongoing developments in the field of track systems, there is still room for further improvements for track systems configured to be used on wheeled vehicles. More particularly, improvements related to characteristics of track system, such as ride quality, traction and durability,

are desirable. In addition, improving such characteristics in a cost effective manner has proven to be challenging, and thus continued improvements in this area remain desirable.

SUMMARY

(5) It is therefore an object of the present technology to ameliorate the situation with respect to at least one of the inconveniences present in the prior art.

(6) In accordance with one aspect of the present technology, there is provided a multi-member frame assembly for a track system, the multi-member frame assembly being operatively connectable to an axle of a vehicle. The multi-member frame assembly includes a pivot pin defining a pivot axis, the pivot pin being at least indirectly connectable to the axle of the vehicle, a leading frame member pivotally connected to the pivot pin for pivoting about the pivot axis, a trailing frame member pivotally connected to the pivot pin for pivoting about the pivot axis, the trailing frame member pivoting independently from the leading frame member, and a resilient member pivotally connected to the leading frame member and to the trailing frame member, the resilient member extending generally vertically between the leading frame member and the trailing frame member.

(7) In some implementations, the leading frame member defines a leading loop structured and dimensioned for receiving the pivot pin therein, and a leading annular spacing is defined between the leading loop and the pivot pin. The trailing frame member defines a trailing loop structured and dimensioned for receiving the pivot pin therein, and a trailing annular spacing is defined between the trailing loop and the pivot pin. The multi-member frame assembly further includes a leading annular torsion spring disposed in the leading annular spacing, the leading annular torsion spring being connected to the leading frame member and to the pivot pin for pivotally biasing the leading frame member about the pivot axis, and a trailing annular torsion spring disposed in the trailing annular spacing, the trailing annular torsion spring being connected to the trailing frame member and to the pivot pin for pivotally biasing the trailing frame member about the pivot axis.

(8) In some implementations, the multi-member frame assembly further includes a leading tab connected to the leading frame member, and a trailing tab connected to the trailing frame member, the leading and trailing tabs extending on a same longitudinal side of the pivot axis, and the resilient member being pivotally connected to the leading and trailing tabs.

(9) In accordance with another aspect of the present technology, there is provided a track system for use with a vehicle having a chassis and an axle extending laterally outwardly from the chassis for driving the track system. The track system includes a multi-member frame assembly operatively connectable to the axle. The multi-member frame assembly includes a pivot pin defining a pivot axis, the pivot pin being at least indirectly connectable to the axle of the vehicle, a leading frame member pivotally connected to the pivot pin for pivoting about the pivot axis, a trailing frame member pivotally connected to the pivot pin for pivoting about the pivot axis, the trailing frame member pivoting independently from the leading frame member, a leading wheel-bearing frame member at least indirectly pivotally connected to the leading frame member, and a trailing wheel-bearing frame member at least indirectly pivotally connected to the trailing frame member, a resilient member interconnecting the leading frame member and the trailing frame member, a leading idler wheel assembly rotatably connected to the leading wheel-bearing frame member, a trailing idler wheel assembly rotatably connected to the trailing wheel-bearing frame member, a first support wheel assembly at least indirectly connected to the leading wheel-bearing frame member, a second support wheel assembly at least indirectly connected to the trailing wheel-bearing frame member, and a sprocket wheel operatively connected to the axle.

(10) In some implementations, the leading frame member defines a leading loop structured and dimensioned for receiving the pivot pin therein, and a leading annular spacing is defined between the leading loop and the pivot pin. The trailing frame member defines a trailing loop structured and dimensioned for receiving the pivot pin therein, and a trailing annular spacing is defined between the trailing loop and the pivot pin. The track system further includes a leading annular torsion

spring disposed in the leading annular spacing, the leading annular torsion spring being connected to the leading frame member and to the pivot pin for pivotally biasing the leading frame member about the pivot axis, and a trailing annular torsion spring disposed in the trailing annular spacing, the trailing annular torsion spring being connected to the trailing frame member and to the pivot pin for pivotally biasing the trailing frame member about the pivot axis.

(11) In some implementations, the leading and trailing annular torsion springs are rubber bushings.

(12) In some implementations, the track system further includes a leading tab connected to the leading frame member, and a trailing tab connected to the trailing frame member, the leading and trailing tabs extending on a same longitudinal side of the pivot axis, and the resilient member being pivotally connected to the leading and trailing tabs.

(13) In some implementations, the resilient member extends generally vertically.

(14) In some implementations, the resilient member is disposed laterally inwardly from a majority of at least one of the leading and trailing frame members.

(15) In some implementations, the resilient member includes first and second resilient elements, and a plate connected to the first and second resilient elements and disposed between the first and second resilient elements.

(16) In some implementations, the track system further includes an axle casing rotatably connectable to the axle, and at least one pivot pin bracket connecting the axle casing to the pivot pin, thereby operatively connecting the multi-member frame assembly to the axle of the vehicle.

(17) In some implementations, the track system further includes leading and trailing stops connected to the axle casing, the leading and trailing stops being structured and dimensioned to limit pivotal movement of the leading and trailing frame members. In some implementations, the track system further includes an endless track extending around the sprocket wheel, the leading idler wheel assembly, the trailing idler wheel assembly, and the first and second support wheel assemblies, the endless track being drivable by the sprocket wheel.

(18) In some implementations, the endless track has a ground-contacting area that increases in size as a load borne by the track system increases.

(19) Implementations of the present technology each have at least one of the above-mentioned object and/or aspects, but do not necessarily have all of them. It should be understood that some aspects of the present technology that have resulted from attempting to attain the above-mentioned object may not satisfy this object and/or may satisfy other objects not specifically recited herein.

(20) Should there be any difference in the definitions of term in this application and the definition of these terms in any document included herein by reference, the terms as defined in the present application take precedence.

(21) Additional and/or alternative features, aspects, and advantages of implementations of the present technology will become apparent from the following description, the accompanying drawings, and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:

(2) FIG. 1 is a perspective view taken from a top, front, right side of a track system being configured to be operatively connected on a left side of a vehicle;

(3) FIG. 2 is a front elevation view of a vehicle having the track system of FIG. 1 operatively connected to the left side thereof, and another track system being a mirror image of the track system of FIG. 1 operatively connected to the right side thereof;

- (4) FIGS. 3A and 3B are right side elevation views of the track system of FIG. 1;
- (5) FIG. 4 is a perspective view taken from a top, front, right side of the track system of FIG. 1, with the endless track removed;
- (6) FIG. 5 is a top plan view of the track system of FIG. 4;
- (7) FIG. 6 is a left side elevation view of the track system of FIG. 4;
- (8) FIG. 7 is a partially exploded, perspective view taken from a top, front, right side of the track system of FIG. 6;
- (9) FIG. 8 is a partially exploded, perspective view taken from a bottom, front, right side of the track system of FIG. 6;
- (10) FIG. 9 is a perspective view taken from a top, front, right side of a leading frame member, a trailing frame member, and a resilient member of the track system of FIG. 6;
- (11) FIG. 10 is a top plan view of the leading frame member, the trailing frame member, and the resilient member of the track system of FIG. 6;
- (12) FIG. 11 is a perspective view taken from a top, front, right side of the leading frame member of the track system of FIG. 6;
- (13) FIG. 12 is a perspective view taken from a bottom, front, right side of the trailing frame member of the track system of FIG. 6; and
- (14) FIG. 13 is an exploded, perspective view taken from a top, front, left side of the resilient member of the track system of FIG. 6.

DETAILED DESCRIPTION

(15) With reference to FIGS. 1 to 8, an implementation of the present technology, track system **40**, will be described. It is to be expressly understood that the track system **40** is merely an implementation of the present technology. Thus, the description thereof that follows is intended to be only a description of illustrative examples of the present technology. This description is not intended to define the scope or set forth the bounds of the present technology. In some cases, what are believed to be helpful examples of modifications or alternatives to track system **40** may also be set forth below. This is done merely as an aid to understanding, and, again, not to define the scope or set forth the bounds of the present technology. These modifications are not an exhaustive list, and, as a person skilled in the art would understand, other modifications are likely possible. Further, where this has not been done (i.e. where no examples of modifications have been set forth), it should not be interpreted that no modifications are possible and/or that what is described is the sole manner of implementing or embodying that element of the present technology. As a person skilled in the art would understand, this is likely not the case. In addition, it is to be understood that the track system **40** may provide in certain aspects a simple implementation of the present technology, and that where such is the case it has been presented in this manner as an aid to understanding. As persons skilled in the art would understand, various implementations of the present technology may be of a greater complexity than what is described herein.

(16) Referring to FIG. 2, the track system **40** is for use with a vehicle **60** (schematically shown in FIG. 2) having a chassis **62** and an axle **68** extending laterally outwardly from the chassis **62** for driving the track system **40**. The chassis **62** supports the various components of the vehicle **60**. In some implementations, the vehicle **60** is an agricultural vehicle. However, the track system **40** could be used on different types of vehicles that serve many different functions. In FIG. 2, a track system **40'** being a mirror image of the track system **40** is shown operatively connected to an axle **68'** extending on a right side of the vehicle **60**. Only the track system **40** configured to be operatively connected to a left side of the vehicle **60** will be described below and in the accompanying Figures.

(17) In the context of the following description, “outward” or “outwardly” means away from a longitudinal center plane **66** (FIG. 2) of the chassis **62** of the vehicle **60**, and “inward” or “inwardly” means toward the longitudinal center plane **66**. In addition, in the context of the following description, “longitudinal” or “longitudinally” means in a direction parallel to the

longitudinal center plane **62** of the chassis **60** in a plane parallel to flat level ground, “lateral” or “laterally” means in a direction perpendicular to the longitudinal center plane **66** in a plane parallel to flat level ground, and “vertical” or “vertically” means in a direction perpendicular to the longitudinal center plane **66** along a height direction of the track system **40** in a plane perpendicular to flat level ground. Note that in the Figures, a “+” symbol is used to indicate an axis of rotation or pivot. In the context of the present technology, the term “axis” may be used to indicate an axis of rotation, or the term may refer to a “pivot joint” that includes all the necessary structure (bearing structures, pins, axles and other components) to permit a structure to pivot about such axis, as the case may be.

(18) Moreover, the direction of forward travel of the track system **40** is indicated by an arrow **80** (FIG. **1**). In the present description, the “leading” components are identified with an “1” added to their reference numeral (i.e. components towards the front of the vehicle defined consistently with the vehicle's forward direction of travel), and the “trailing” components are identified with a “t” added to their reference numeral (i.e. components towards the rear of the vehicle defined consistently with the vehicle's forward direction of travel).

(19) Furthermore, it is to be understood in the present description that a wheel assembly includes one or more wheels, an axle for supporting the one or more wheels, and the components (bearings, seals, etc.) that are necessary for the wheel(s) to rotate. As such, the different wheel assemblies will not be described in great details in the current description. Moreover, the expression “at least indirectly connected” is understood to mean that a component may be connected to another component via one or more intermediate structures or members, and that these intermediate structures are not necessarily described in the current description. Finally, the expression “track system is at a rest position” is understood to mean that the track system **40** is connected to the vehicle **60**, is stationary, and is disposed on flat, level ground.

(20) Referring to FIGS. **1** to **8**, the track system **40** will be generally described. The track system **40** has a multi-member frame assembly **100**. The multi-member frame assembly **100** includes a pivot pin **110** defining a pivot axis **112**, a leading frame member **1201** pivotally connected to the chassis **62** of the vehicle **60** via the pivot pin **110** for pivoting about the pivot axis **112**, and a trailing frame member **120t** also pivotally connected to the pivot pin **110** for pivoting about the pivot axis **112** independently from the leading frame member **1201**.

(21) The track system **40** further has an axle casing **130** defining an aperture **132** that is sized and configured for allowing passage of the axle **68** therethrough. The axle casing **130** is rotatably connected to the axle **68** via bearings (not shown) disposed in the aperture **132**. Inward and outward pivot pin brackets **140** (shown in FIG. **7**) are connected to the axle casing **130**, and extend downwardly therefrom. In the present implementation, the pivot pin brackets **140** are pillow blocks, but they could differ in other implementations. The inward and outward pivot pin brackets **140** are operatively connected to the inward and outward ends of the pivot pin **110**. The multi-member frame assembly **100** is thus operatively connected to the axle **68** of the vehicle **60** via the pivot pin brackets **140**. Leading and trailing stops **1341**, **134t** (FIGS. **7** and **8**) are connected to the axle casing **130**. The leading and trailing stops **1341**, **134t** are structured and dimensioned to limit pivotal movement of the leading and trailing frame members **1201**, **120t**. It is contemplated that, in some implementations, the leading and trailing stops **1341**, **134t** could be connected to the chassis **62** of the vehicle **60** rather than connected to the axle casing **130**.

(22) The multi-member frame assembly **100** further includes a leading wheel-bearing frame member **1601** pivotally connected to the leading frame member **1201**, and a trailing wheel-bearing frame member **160t** pivotally connected to the trailing frame member **120t**. Pins **162** (FIGS. **7** and **8**) are used to pivotally connect the leading wheel-bearing frame member **1601** and the trailing wheel-bearing frame member **160t** to the leading frame member **1201** and the trailing frame member **120t** respectively. A trailing tandem assembly **170t** is pivotally connected to the trailing wheel-bearing frame member **120t**.

(23) The track system **40** further includes a resilient member **200** interconnecting the leading frame member **1201** and the trailing frame member **120t**. As such, the pivotal motion of the leading frame member **1201** relative to the trailing frame member **120t** is dampened by the resilient member **200**.

(24) When the track system **40** supports the weight of the vehicle **60**, the resilient member **200** is deformed (i.e. compressed). Under certain conditions, vibrations that are caused by the ground surface on which the track system **40** travels, and that are transferred to the leading and trailing frame members **1201**, **120t**, are dampened by the resilient member **200**, and thus shocks and vibrations transferred from the ground to the vehicle **60** are attenuated in certain conditions. As a result, wear of components is reduced and the comfort that a user of the vehicle **60** experiences when riding the vehicle **60** equipped with the track system **40** is improved compared to other track systems.

(25) A leading idler wheel assembly **4001** is rotatably connected to the leading wheel-bearing frame member **1601**, and a trailing idler wheel assembly **400t** is rotatably connected to the trailing wheel-bearing frame member **160t**. A plurality of support wheel assemblies **410a**, **410b**, **410c** are disposed intermediate the leading idler wheel assembly **4001** and the trailing idler wheel assembly **400t**. The support wheel assemblies **410b**, **410c** are rotatably connected to the trailing tandem assembly **170t**, and as mentioned above, the trailing tandem assembly **170t** is pivotally connected to the trailing wheel-bearing frame member **120t**. It is contemplated that the support wheel assemblies **410a**, **410b**, and **410c** could be connected in different configurations in other implementations.

(26) Referring to FIG. 2, the track system **40** includes a gearbox **500** operatively connected to the axle **68** of the vehicle **60**, and supported by the axle casing **130**. The gearbox **500** is adapted to modify the driving force provided by the axle **68**. The track system **40** further includes a sprocket wheel **550** operatively connected to the axle the axle **68**. As such, the driving force is transferred from the axle **68** to the sprocket wheel **550**. In other implementations, the gearbox **500** could be omitted and the axle **68** could be directly operatively connected to the sprocket wheel **550** and extending through the axle casing **130**.

(27) The track system **40** further includes an endless track **600** extending around the sprocket wheel **550**, the leading idler wheel assembly **4001**, the trailing idler wheel assembly **400t**, and the plurality of support wheel assemblies **410a**, **410b**, **410c**. The endless track **600** is drivable by the sprocket wheel **550**. The leading idler wheel assembly **4001** is operatively connected to a tensioner assembly **402** (FIG. 7) adapted to move the leading idler wheel assembly **4001** forward and backward to control a tension in the endless track **600**. In some implementations, the tensioner assembly **402** includes a dynamic tensioning device as described in commonly owned International Patent Application No. PCT/CA2016/050419, filed Apr. 11, 2016, entitled “Dynamic Tensioner Locking Device for a Track System and Method Thereof”, and published as WO 2016/161528. The content of this application is incorporated herein by reference in its entirety.

(28) Referring back to FIGS. 1, 3A and 3B, the endless track **600** is an endless polymeric track. The endless track **600** has an inner surface **602** engaging the leading idler wheel assembly **4001**, the trailing idler wheel assembly **400t**, and the plurality of support wheel assemblies **410a**, **410b**, **410c**. Lugs **604** are disposed on a central portion of the inner surface **602** and are engageable by the sprocket wheel **550**. As such, the track system **40** is a “positive drive” track system. Friction drive track systems are also contemplated as being an alternative to the present implementations. Moreover, hybrid drive track systems (i.e. the sprocket wheel **550** drives the endless track **600** via friction and driving engagement of lugs **604**) are also contemplated. The idler and support wheel assemblies **4001**, **400t**, **410a**, **410b**, **410c** have laterally spaced-apart wheels engaging the inner surface **602** of the endless track **600** on either side of the lugs **604**. The endless track **600** also has an outer surface **606** with a tread **608** (FIGS. 1, 3A and 3B) selected for ground engagement. The tread **608** varies in different implementations according to the type of vehicle on which the track system **40** is to be used with and/or the type of ground surface on which the vehicle is destined to travel. It is contemplated that within the scope of the present technology, the endless track **600** may

be constructed of a wide variety of materials and structures including metallic components known in track systems. Referring to FIGS. 3A and 3B, the endless track **600** has a leading segment **610**, a ground engaging segment **620** and a trailing segment **630**. The generally triangular shape of the track system **40** causes the endless track **600** to have the segments **610**, **620**, **630**, but as other configurations of the track system **40** are contemplated, the endless track **600** could have more or less segments in other implementations.

(29) Referring now to FIGS. 7 and 8, the leading and trailing frame members **1201**, **120t** will be described. The leading and trailing frame members **1201**, **120** are pivotally connected to and supported by the pivot pin **110**. The leading and trailing frame members **1201**, **120t** have apertures defined by leading and trailing loops **1801**, **180t** respectively. The pivot pin **110** extends through the apertures of the loops **1801**, **180t** similar to a pin in a hinge assembly, and enables the pivoting of the leading and trailing frame members **1201**, **120t** about the pivot axis **112**.

(30) Referring to FIGS. 3A, 3B, 7 and 8, the leading and trailing frame members **120**, **120t** of the track system **40** define a somewhat scissor-like structure, with each frame member **1201**, **120t** pivoting about the pivot pin **110**, and the resilient member **200** interconnected therebetween. Each one of the leading and trailing wheel-bearing frame members **1601**, **160t**, is in turn pivotally connected to the leading and trailing frame members **1201**, **120t**, respectively via the pins **162**. The pivoting of each of these structures may assist in reducing the vertical displacements and vibrations transferred from the track system **40** to the chassis **62** of the vehicle **60** under certain conditions. In some implementations, the track system **40** further includes bushing assemblies operatively connected between the axle assemblies rotatably connecting the wheel assemblies **4001**, **400t**, **410a**, **410b**, **410c** to their corresponding component of the frame assembly **100**. The bushing assemblies further assist in reducing the vibrations transferred from the track system **40** to the chassis **62** of the vehicle **60** under certain conditions.

(31) In addition, having the track system **40** with such a scissor-like structure has other advantages in certain situations. For example, as the weight of the vehicle **60** increases, for example during harvesting or loading operations, the scissor-like structure can open and a ground-contacting portion of the endless track **600** occurs over an increased surface area (i.e. the ground engaging segment **620** increases in size as the load borne by the track system **40** increases—at least for some increases in load—depending on the design of a specific track system). As a result, in some circumstances, the pressure applied to the ground by the endless track **600** (owing to the weight and load of the vehicle **60**) increases at a lower rate than the weight of the vehicle **60**. In certain implementations, this will allow the track system **40** to bear additional loads as compared with conventional track systems.

(32) Referring back to FIGS. 7 and 8, a leading annular spacing **1821** is defined between the leading loop **1801** and the pivot pin **110**. Similarly, a trailing annular spacing **182t** is defined between the trailing loop **180t** and the pivot pin **110**. In the present implementation, the track system **40** further includes a leading annular torsion spring **1841** disposed in the leading annular spacing **1821**. The leading annular torsion spring **1841** is connected to the leading frame member **1201** and to the pivot pin **110** for pivotally biasing the leading frame member **1201** about the pivot axis **112**. Similarly, a trailing annular torsion spring **184t** is disposed in the trailing annular spacing **182t**. The trailing annular torsion spring **184t** is connected to the trailing frame member **120t** and the pivot pin **110** for pivotally biasing the trailing frame member **120t** about the pivot axis **112**. In the present implementation, the leading and trailing annular torsion springs **1841**, **184t** are rubber bushings, but could be configured otherwise in other implementations. The leading and trailing annular torsion springs **1841**, **184t** not only bias the leading and trailing frame members **1201**, **120t** towards a rest position, but also limit the entry of dirt, debris and mud in the annular spacings **1821**, **182t** that could impair the pivotal connection of the frame members **1201**, **120t**. In some implementations, the leading annular torsion spring **1841** and the trailing annular torsion spring **184t** are integrally formed and define an annular torsion spring extending in both the leading and

trailing annular spacings **1821**, **182t**.

(33) In some implementations, the track system **40** includes bearings (not shown) disposed in the leading annular spacing **1821** and in the trailing annular spacing **182t**. The bearings are rotationally connecting the leading frame member **1201** and the trailing frame member **120t** to the pivot pin **110**. The presence of the annular torsion springs **1841**, **184t** in such implementations can increase the durability of the bearings as entry of dirt, debris and mud in the annular spacings **1821**, **182t** is limited by the annular torsion springs **1841**, **184t**.

(34) Referring to FIGS. 7 to 12, the leading frame member **1201** has a leading tab **1901** connected thereto and extending on the longitudinal side of the pivot axis **112** that contains the majority of the leading frame member **1201** (i.e. longitudinally at the front of the pivot axis **112**). The trailing frame member **120t** has a trailing tab **190t** connected thereto. The trailing tab **190t** projects from the trailing loop **180t** and extends on the longitudinal side of the pivot axis **112** that contains the majority of the leading frame member **1201** (i.e. longitudinally at the front of the pivot axis **112**). The trailing tab **190t** extends above the leading tab **1901**. The resilient member **200** is pivotally connected to the leading and trailing tabs **1901**, **190t** using pins **192**. The pivot connection of the resilient member **200** to the leading and trailing frame members **1201**, **120t** causes that the compression stresses that are applied to the resilient member **200** are along an axis **194** (FIGS. 3A, 3B, 9 and 10) extending between the center of the pins **192**. The compression stresses are axially applied along the axis **194** and as such, the resilient member **200** is subjected to no or little shear stresses. This feature can enhance the durability of the resilient member **200** in certain conditions as the materials forming the resilient member **200** work best when subjected to axial compression stresses. As best seen in FIG. 9, when the resilient member **200** is interconnected between the leading and trailing frame members **1201**, **120t** via the tabs **1901**, **190t**, the resilient member **200** extends generally vertically. The axis **194** thus extends generally vertically, as seen in FIGS. 3A, 3B, 9 and 10. The expression “generally vertically” means that the resilient member **200** extends in a direction perpendicular to the longitudinal center plane **66** along a height direction of the track system **40** in a plane perpendicular to flat level ground, and within ± 30 degrees from the vertical direction. In addition, the resilient member **200** is disposed laterally inwardly from a majority of the leading frame member **1201**.

(35) This positioning of the resilient member **200** has been found to improve to overall packaging of the track system **40**. In addition, the resilient member **200** is partially protected from dirt, mud and debris projections by the leading frame member **1201**. Furthermore, it has been found that having the resilient member **200** extending generally vertically reduces mud accumulation thereon over time compared to other track systems having a resilient member that extends otherwise. For example, in track systems having a resilient member extending horizontally, mud can pile up on surfaces of the resilient member extending horizontally and can become compact and hard over time, at least in certain conditions. The accumulated mud may impair the action the resilient member, and reduce its durability in certain conditions. In contrast, in the present implementation, the resilient member **200** extends generally vertically, and mud has less horizontal surface to accumulate on. Even if mud had accumulated on the upper end of the resilient member **200**, the action and durability thereof would not be significantly affected. In other implementations, it is contemplated that the resilient member **200** could be disposed at the rear of the pivot axis **112**, and be protected from dirt, mud and debris projections by the trailing frame member **120t**.

(36) Moreover, having the annular torsion springs **1821**, **182t** used in combination with the resilient member **200** has been found to provide improved damping of the vibrations induced in the track system **40** when in use, and provide a relatively low-cost solution to enhance the ride comfort on a track system **40** having a scissor-like structure. This feature also permits that, in some implementations, the resilient member **200** be of a smaller size compared to a track system **40** that would not include the annular torsion springs **1821**, **182t**, which further improves the overall packaging of the track system **40** and further reduces the possibilities for mud accumulation related

problems.

(37) Furthermore and referring to FIG. 13, the resilient member **200** is seen to include three resilient elements **202**, and two plates **204** connected to the resilient elements **202** and disposed between the adjacent resilient elements **202**. The resilient elements **202** are made of a rubber-based material, and the plates **204** are made of a metallic material, such as steel. It is contemplated that the plates **204** could be made of a different material, such as ultra-high molecular weight polymeric material (UHMW), or aluminium, in other implementations. The plates **204** reduce the undesired deformation of the resilient elements **202** in the lateral direction due to shear stress. The plates **204** thus increase the durability of the resilient elements **202** at least in certain conditions. The resilient member **200** could be structured otherwise in other implementations. For example, the resilient member **200** could include a coil spring, a shock absorber, a fluid-based suspension element, etc.

(38) The multi-member frame assembly **100** and the track system **40** implemented in accordance with some non-limiting implementations of the present technology can be represented as presented in the following numbered clauses.

(39) Modifications and improvements to the above-described implementations of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting. The scope of the present technology is therefore intended to be limited solely by the scope of the appended claims.

Claims

1. A multi-member frame assembly for a track system, the multi-member frame assembly being operatively connectable to an axle of a vehicle, the multi-member frame assembly comprising: a pivot pin defining a pivot axis, the pivot pin being at least indirectly connectable to the axle of the vehicle; a leading frame member pivotally connected to the pivot pin for pivoting about the pivot axis; a trailing frame member pivotally connected to the pivot pin for pivoting about the pivot axis, the trailing frame member pivoting independently from the leading frame member; and a resilient member pivotally connected to the leading frame member and to the trailing frame member, the resilient member extending generally vertically between the leading frame member and the trailing frame member.
2. The multi-member frame assembly of claim 1, wherein: the leading frame member defines a leading loop structured and dimensioned for receiving the pivot pin therein, and a leading annular spacing is defined between the leading loop and the pivot pin; the trailing frame member defines a trailing loop structured and dimensioned for receiving the pivot pin therein, and a trailing annular spacing is defined between the trailing loop and the pivot pin; and the multi-member frame assembly further includes: a leading annular torsion spring disposed in the leading annular spacing, the leading annular torsion spring being connected to the leading frame member and to the pivot pin for pivotally biasing the leading frame member about the pivot axis; and a trailing annular torsion spring disposed in the trailing annular spacing, the trailing annular torsion spring being connected to the trailing frame member and to the pivot pin for pivotally biasing the trailing frame member about the pivot axis.
3. The multi-member frame assembly of claim 1, further comprising a leading tab connected to the leading frame member, and a trailing tab connected to the trailing frame member, the leading and trailing tabs extending on a same longitudinal side of the pivot axis, and the resilient member being pivotally connected to the leading and trailing tabs.
4. A track system for use with a vehicle having a chassis and an axle extending laterally outwardly from the chassis for driving the track system, the track system comprising: a multi-member frame assembly operatively connectable to the axle, the multi-member frame assembly including: a pivot pin defining a pivot axis, the pivot pin being at least indirectly connectable to the axle of the vehicle; a leading frame member pivotally connected to the pivot pin for pivoting about the pivot

axis; a trailing frame member pivotally connected to the pivot pin for pivoting about the pivot axis, the trailing frame member pivoting independently from the leading frame member; a leading wheel-bearing frame member at least indirectly pivotally connected to the leading frame member; and a trailing wheel-bearing frame member at least indirectly pivotally connected to the trailing frame member; a resilient member interconnecting the leading frame member and the trailing frame member; a leading idler wheel assembly rotatably connected to the leading wheel-bearing frame member; a trailing idler wheel assembly rotatably connected to the trailing wheel-bearing frame member; a first support wheel assembly at least indirectly connected to the leading wheel-bearing frame member; a second support wheel assembly at least indirectly connected to the trailing wheel-bearing frame member; and a sprocket wheel operatively connected to the axle.

5. The track system of claim 4, wherein: the leading frame member defines a leading loop structured and dimensioned for receiving the pivot pin therein, and a leading annular spacing is defined between the leading loop and the pivot pin; the trailing frame member defines a trailing loop structured and dimensioned for receiving the pivot pin therein, and a trailing annular spacing is defined between the trailing loop and the pivot pin; and the track system further includes: a leading annular torsion spring disposed in the leading annular spacing, the leading annular torsion spring being connected to the leading frame member and to the pivot pin for pivotally biasing the leading frame member about the pivot axis; and a trailing annular torsion spring disposed in the trailing annular spacing, the trailing annular torsion spring being connected to the trailing frame member and to the pivot pin for pivotally biasing the trailing frame member about the pivot axis.

6. The track system of claim 5, wherein the leading and trailing annular torsion springs are rubber bushings.

7. The track system of claim 6, further comprising a leading tab connected to the leading frame member, and a trailing tab connected to the trailing frame member, the leading and trailing tabs extending on a same longitudinal side of the pivot axis, and the resilient member being pivotally connected to the leading and trailing tabs.

8. The track system of claim 7, wherein the resilient member extends generally vertically.

9. The track system of claim 8, wherein the resilient member is disposed laterally inwardly from a majority of at least one of the leading and trailing frame members.

10. The track system of claim 4, wherein the resilient member includes first and second resilient elements, and a plate connected to the first and second resilient elements and disposed between the first and second resilient elements.

11. The track system of claim 4, further comprising: an axle casing rotatably connectable to the axle; and at least one pivot pin bracket connecting the axle casing to the pivot pin, thereby operatively connecting the multi-member frame assembly to the axle of the vehicle.

12. The track system of claim 11, further comprising leading and trailing stops connected to the axle casing, the leading and trailing stops being structured and dimensioned to limit pivotal movement of the leading and trailing frame members.

13. The track system of claim 4, further comprising an endless track extending around the sprocket wheel, the leading idler wheel assembly, the trailing idler wheel assembly, and the first and second support wheel assemblies, the endless track being drivable by the sprocket wheel.

14. The track system of claim 13, wherein the endless track has a ground-contacting area that increases in size as a load borne by the track system increases.
